ACS Catalysis

pubs.acs.org/acscatalysis

Research Article

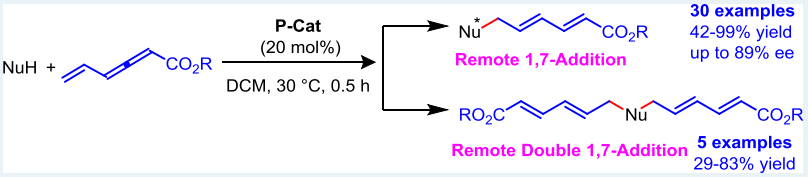
Phosphine-Catalyzed Remote 1,7-Addition for Synthesis of Diene Carboxylates

Jiaxu Feng and You Huang*

Cite This: *ACS Catal.* 2020, 10, 3541−3547

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: A phosphine-catalyzed remote 1,7-addition of vinyl allenoates has been developed, providing a series of 1,3-dienes derivatives in high yields (up to 99%) and with good chemo-, regio-, and stereoselectivity. This reaction demonstrated that the introduction of vinyl in allenoates effectively extended reaction types of phosphine-catalyzed nucleophilic addition of allenoates, leading to concise synthesis of diene carboxylates. Notably, the enantioselective variant of this 1,7-addition can also be performed by chiral phosphine catalyst.

KEYWORDS: *γ-vinyl allenoates, phosphine, catalysis, remote 1,7-addition, synthesis, selectivity, dienes carboxylates*

Downloaded via FUDAN UNIV on June 24, 2020 at 00:35:27 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Diene carboxylates are crucial skeletons which not only widely exist in numerous natural products and bioactive compounds[1] such as bacteriostatic antibiotic Macrolactin A,[2] vitro antifugual activity (+)-papulacandin D (Figure 1),[3] but

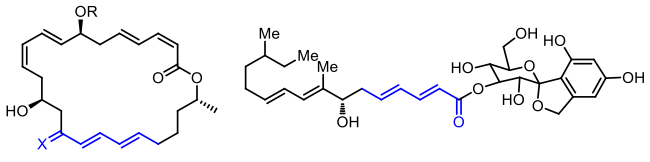


Figure 1. Bioactive compounds containing diene carboxylates.

also used as versatile building blocks in a number of organic transformations.[4] Although many methods have been developed for their synthesis including transition-metal-catalyzed coupling,[5] Wittig reaction,[6] isomerization of allenes,[7] and others,[8] most of them suffered some undesired limitations such as contamination of transition metals, stoichiometric amount of byproducts, and sometimes long steps. Thereby, the development of metal-free, catalytic, and concise methods for the synthesis of diene carboxylates is still in high demand.

Phosphine-catalyzed reactions have received wide attention during past decades, because they provide a powerful and metal-free access to various C−C bonds formation in organic synthesis, eliminating the contamination of metals in organic products.[9,10] Major contributions in this field focused on electrophilic additions, in which the in situ formed zwitterionic intermediate acts as a nucleophile to react with a certain electrophile.[11] Later on, nucleophilic addition was found to be feasible when appropriate pronucleophiles were used, affording a new protocol for C−C bonds formation.[12] The pioneering γ-umpolung additions of pronucleophiles to allenoates were developed independently by Trost and Lu in the 1990s (Scheme 1a).[13,14]

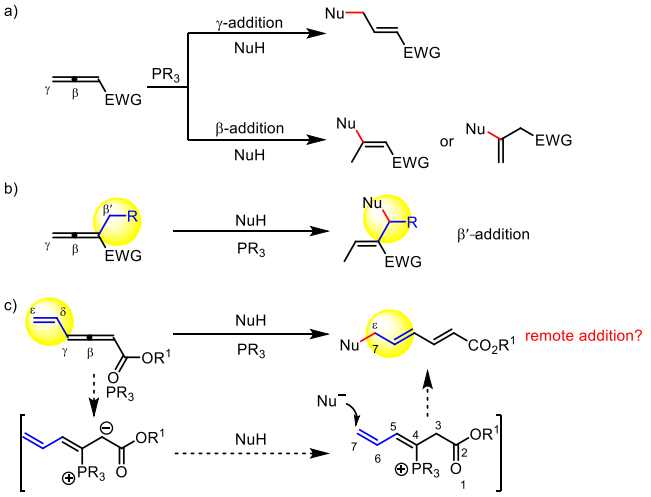
Later on, Huang and Lu's group reported a new β-addition of allenoates.[15] Despite these advances, more variations on allenoates are still needed for the development of new types of additions and versatile product motifs. In this context, Kwon and co-workers introduced α-alkyl into allenoates, and achieved a novel β′-umpolung addition for the synthesis of functionalized acrylates (Scheme 1b).[16] In 2009, we found the new reaction site of δ-carbon of γ-methyl-substituted allenoates first,[17a] then a series of phosphine-catalyzed annulation reactions were developed based on this discovery. Very recently, we designed a new kind δ-sulfonamido-allenoates to develop a sequential [3 + 3]/[3 + 3] annulation of allenoates;[17f] to continue our interests in phosphine-catalyzed reactions,[17] we envisioned a new variation of allenoates for the synthesis of diene carboxylates: installing a γ-vinyl group on allenoates to extend the π-system of the zwitterionic intermediate to remote position, thus allowing to construct a diene carboxylate motif (Scheme 1c).

Received: February 4, 2020
Revised: February 18, 2020
Published: February 19, 2020

© 2020 American Chemical Society

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547

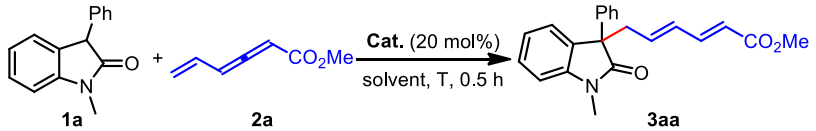
Scheme 1. Phosphine-Catalyzed Addition of Activated Allenes[a]



[a](a) Two common additions of simple activated allenes. (b) Reported variation: alkylated activated allenes for β'-addition. (c) Our proposed variation: vinylated activated allenes for remote 1,7-addition.

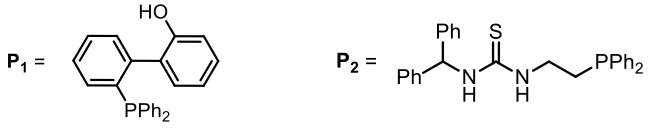
However, to enable this new reaction pattern, the following challenges need to be considered: (1) regioselectivity control on multiple reaction-sites such as nucleophilic α and γ positions, electrophilic γ and ε positions after H abstraction. (2) Appropriate pronucleophiles that enable both H abstraction and subsequent addition step. (3) Difficult stereoselectivity control of remote addition. With these concerns in mind, we commenced our study by selecting 1-methyl-3-phenylindolin-2-one (**1a**) and methyl hexa-2,3,5-trienoate (**2a**) as model substrates under mild conditions (Table 1). When the reactions of **1a** and **2a** were performed in $CHCl_3$ at 30 °C, weakly nucleophilic phosphines such as PPh_3 and other triaryl phosphines showed good reactivity, providing the desired product **3a** in 48−63% yields (Table 1, entries 1−3), When the diarylalkylphosphines Ph_2PEt was used, the reaction gives 56% yield (Table 1, entry 4), whereas strongly nucleophilic phosphine, such as tributyl phosphine, significantly decreased the yield (Table 1, entry 5). Then, we envisioned using bifunctional phosphines bearing additional functional groups to enhance the reactivity; thus, additional interactions between phosphines and the substrates could accelerate the reaction rate. Following this idea, we screened two bifunctional phosphines (entries 6, 7), and ultimately, thiourea-derived phosphine $\mathbf{P_2}$ was found to be the optimal catalyst, affording the desired product with significantly improved yield under the same conditions (77%, entry 7). With $\mathbf{P_2}$ as the catalyst, other reaction parameters such as solvents (entries 8−12), temperatures (entries 13 and 14) were systematically investigated, revealing that the use of DCM as the solvent at 30 °C gave the highest yield (98%, entry 10). When reducing the loading of catalyst to 10 mol %, a lower yield was obtained (86%, entry 15). Meanwhile, a control experiment showed that without the catalyst, no product was afforded and the starting material **1a** was recovered in 90% yield (entry 16). This result indicated that the phosphine catalyst was essential. In this reaction, the thiourea group of the bifunctional phosphines catalyst may play

Table 1. Optimization of Reaction Conditions[a]



entry	cat.	solvent	temp. (°C)	yield (%)[b]
1	PPh_3	$CHCl_3$	30	63
2	$(p\text{-}FC_6H_4)_3P$	$CHCl_3$	30	48
3	$(p\text{-}MeOC_6H_4)_3P$	$CHCl_3$	30	66
4	$(C_6H_5)_2PEt$	$CHCl_3$	30	56
5	PBu_3	$CHCl_3$	30	23
6	P_1	$CHCl_3$	30	57
7	P_2	$CHCl_3$	30	77
8	P_2	THF	30	92
9	P_2	toluene	30	97
10	P_2	DCM	30	98
11	P_2	DCE	30	89
12	P_2	^{i}PrOH	30	89
13	P_2	DCM	20	81
14	P_2	DCM	10	72
15[c]	P_2	DCM	30	86
16		DCM	30	

[a]



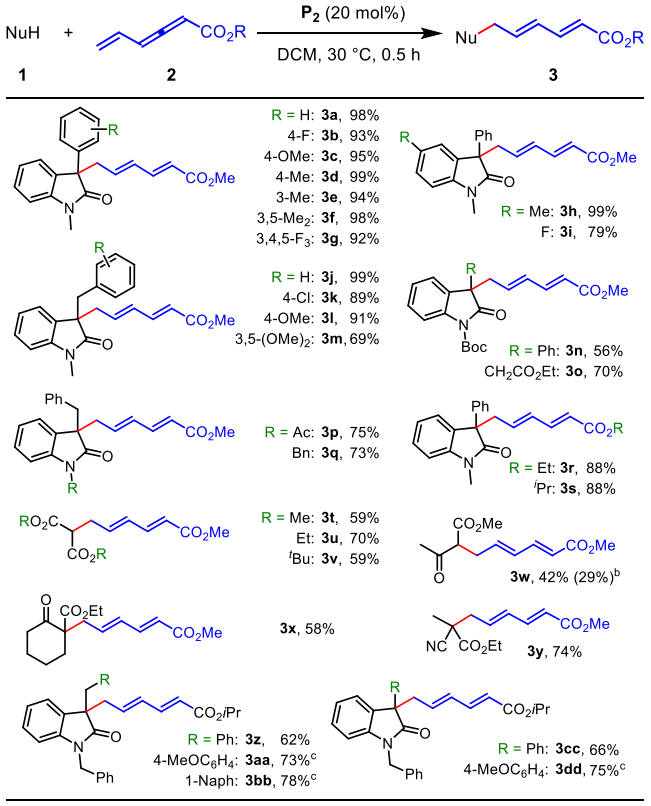
Reaction conditions: **1a** (0.1 mmol) and **2a** (0.3 mmol) in solvent (1.0 mL). [b]Yields of isolated products. [c]10 mol % catalyst was used.

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547

an important role in coordinating the nucleophile to accelerate the reaction.

With the optimal reaction conditions in hand, we turned to examine the substrates scope by using various nucleophiles and vinyl allenoates (Table 2). We were pleased to find that a wide

Table 2. Phosphine-Catalyzed Remote 1,7-Addition[a]



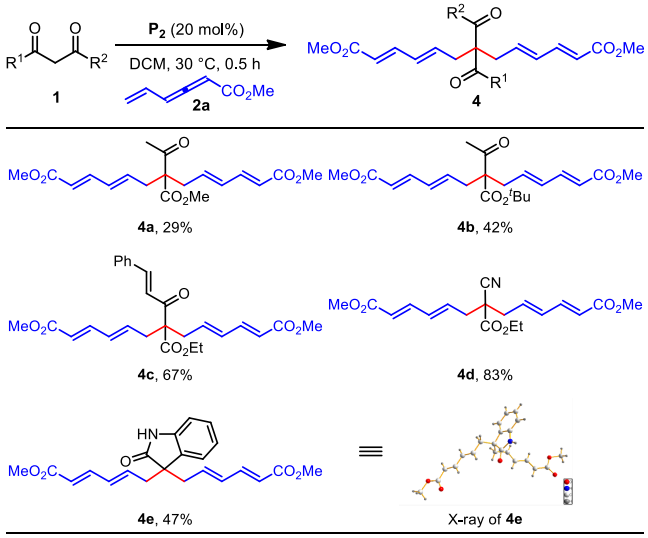
[a]Reaction conditions: 1 (0.1 mmol), 2 (0.3 mmol) in DCM (1.0 mL) at 30 °C; yields of isolated products. [b]Yield of the double addition product in parentheses. [c]Reaction conditions: 1 (0.1 mmol), 2 (0.3 mmol) in toluene (1.0 mL) at 100 °C; yields of isolated products.

array of commercially available oxindole derivatives could react efficiently under the optimized conditions, affording the desired diene carboxylates **3** in good yields (up to 99%). Different substituents on the aryl rings at the 3-position of oxindoles were examined, and the results showed that both electron-withdrawing (**3b** and **3g**) and electron-donating (**3c**, **3d**, **3e**, and **3f**) substituents at the para- and meta-position were tolerated, affording the products in good yields (93−99%). In addition, the electron-donating group (Me) on the C-5 position of the oxindole was also tolerated, providing diene carboxylate **3h** in 99% yield. In contrast, the electron-withdrawing group (F) on the same position led to 79% yield (**3i**), suggesting that electronic effect on the conjugated phenyl ring of oxindole was more significant. Different 3-benzyl-substituted oxindole derivatives also worked well in the reaction and afforded the corresponding products of **3i**, **3j**, and **3l** in good to excellent yields (89−99%); only the 3,5-dimethoxybenzyl-substituted oxindole led to 69% yield (**3m**). All of the products give the *E*,*E*-diene carboxylates that are highly stereoselective, and the structure of **3l** was unequivocally confirmed by single-crystal X-ray analysis (see the Supporting Information).[18]

The protecting group on the nitrogen atom of oxindole can also be replaced by the *tert*-butoxycarbonyl (**3n** and **3o**), acetyl (**3p**), benzyl (**3q**, **3z**, and **3cc**) groups, and the desired products were obtained in 56−75% yields. With oxindoles as pronucleophiles, we next explored the generality of γ-vinyl allenoates, and the results showed that the variation of ester groups were well tolerated, affording the desired dienes carboxylates **3r** and **3s** in the same 88% yield. Besides 3-substituted oxindoles, other pronucleophiles were also compatible with the current method. For example, a variety of 1,3-dicarbonyl substrates afforded the corresponding products in 59−70% yields (**3t**, **3u**, and **3v**). Very interestingly, at the same reaction condition, when using methyl acetoacetate bearing two α-hydrogens as the pronucleophile, apart from that a single addition product can be obtained in 42% yield, a double addition product can also be produced in 29% yield (**3w**). In addition, other cyclic and linear 1,3-dicarbonyl substrates were also successfully compatible, providing single isomers in 58 and 74% yields, respectively (**3x** and **3y**). It should be noteworthy that **3aa**, **3bb**, and **3dd** were produced in toluene at 100 °C.

Inspired by the double addition result of methyl acetoacetate, we next investigated double addition of 1,3-dicarbonyl compounds bearing two α-hydrogens with allenoates to form bis(diene carboxylates) containing an all-carbon quaternary center (Table 3). Because of high instability, these compounds

Table 3. Phosphine-Catalyzed Double 1,7-Addition[a]



[a]Reaction conditions: 1 (0.1 mmol), 2 (0.3 mmol) in DCM (1.0 mL) at 30 °C; yields of isolated products.

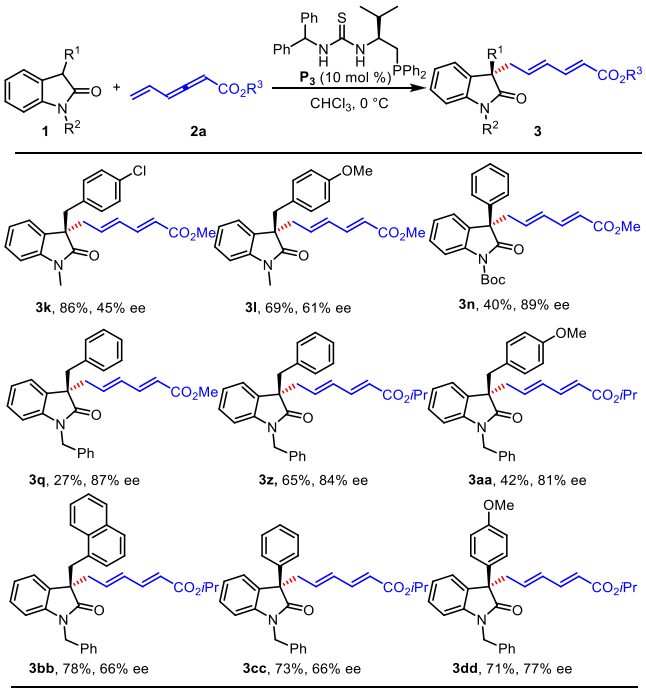
are difficult to be prepared by other methods. Pleasingly, γ-vinyl allenoates reacted with methyl 3-oxobutanoate and *tert*-butyl 3-oxobutanoate, successfully providing the corresponding double remote 1,7-addition products in 29 and 42% yields respectively (**4a** and **4b**). In addition, the Nazarov reagent has been proved to be a suitable substrate, affording a corresponding product in 67% yield (**4c**). Ethyl 2-cyanoacetate was also compatible under the reaction conditions, giving the desired product in 83% yield (**4d**). Notably, indolin-2-one was also accommodated in this reaction, albeit with moderate yield (**4e**). The structure of **4e** was unambiguously confirmed by X-ray single-crystal analysis.[18]

We also tried to develop an asymmetric version of this remote addition reaction. Although many chiral phosphines have been checked, most of them led to either low yields or low ee values (see the Supporting Information).[19] Only chiral phosphine **P**$_3$

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547

gave reasonable results (Table 4). With P_3 as the catalyst, product **3n** can be obtained in 40% yield with good

Table 4. Preliminary Asymmetric Phosphine-Catalyzed 1,7-Addition[a]



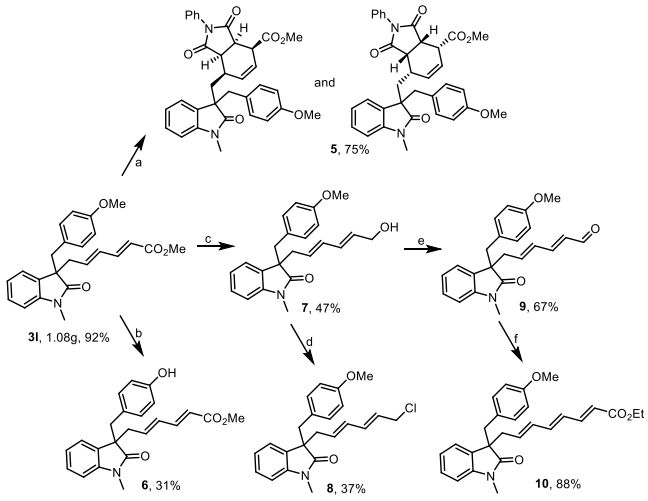
[a]Reaction conditions: 1 (0.1 mmol), 2 (0.3 mmol) in $CHCl_3$ (1.0 mL) at 0 °C; yields of isolated products; ee was determined by chiral high-performance liquid chromatography.

enantioselectivity (89% ee). Other substrates led to good yields and moderate ee values (**3k** and **3l**). When using *N*-benzyl substituted oxindole that served as pronucleophile, different results were obtained when reacted with different allenoates (**3q** and **3z**). This showcased that the steric effects might promote this asymmetric reaction. Moreover, high enantioselectivites were obtained for many products (**3aa**−**3dd**).

To demonstrate the utility of the reaction, gram−scale reaction was conducted, affording **3l** in 92% yield, without the loss of efficiency (1.08 g, Scheme 2). To elaborate the potential synthetic utility of the present protocol, several transformation of diene carboxylates **3l** have been carried out (Scheme 2). Upon treatment with *N*-phenylmaleimide, a Diels−Alder cycloaddition product **5** was afforded in 75% yield. The phenol derivative **6** was prepared through a demethylation process. Additionally, direct reduction of diene carboxylates **3l** led to the formation of allyl alcohol derivative **7**, which underwent oxidation reaction to give the corresponding aldehyde **9**; the aldehyde **9** can be converted to the triene derivative **10** by a Witting reaction. Moreover, allyl chloride derivative **8** was produced through Appel reaction, leaving ample room for further application.

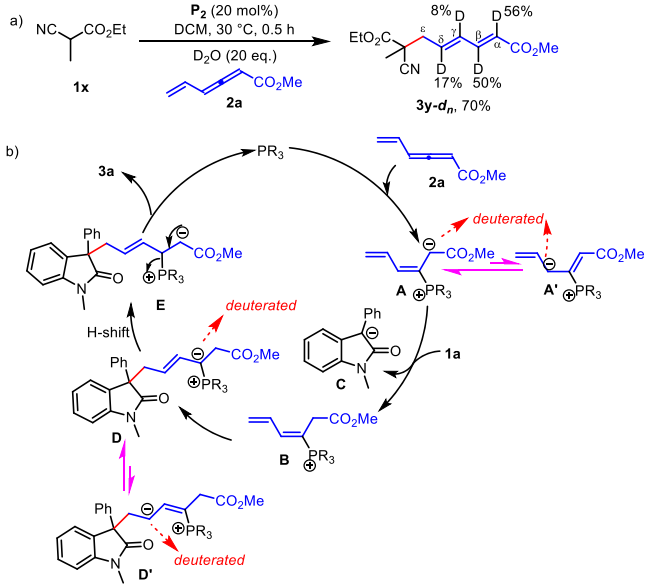
A deuterium labeling experiment was carried out to gain insight into the mechanism. The result showed that α and β positions can be deuterated in almost the same ratio of 56 and 50% respectively, whereas the ratios at γ and δ position were less than 20%, which suggests that α and β positions in the intermediate may display stronger nucleophilicity (Scheme 3a). Based on these results, a possible mechanism for this remote 1,7-

Scheme 2. Gram-Scale Reaction and Synthetic applications[a]



[a]*N*-Phenylmaleimide (1.1 equiv), toluene, 120 °C, 36 h. [b]BBr_3 (8.0 equiv), DCM, −20 °C to rt, 1 h. [c]$liAlH_4$ (3.0 equiv), tetrahydrofuran, rt, 10 min. [d]PPh3 (1.5 equiv), DCM and CCl_4, 0 °C to rt, 12 h. [e]MnO_2 (21.3 equiv), EA, reflux, 7 h. [f]Wittig reagent (2.0 equiv), 4 Å MS, DCM, 50 °C, 24 h.

Scheme 3. Deuterium-Labeling Experiment and Proposed Mechanism[a]



[a](a) Deuterium-labeling experiment. (b) Proposed mechanism.

addition was proposed (Scheme 3b). First, nucleophilic addition of phosphine catalyst to the γ-vinyl substituted allenoate **2a** formed zwitterion **A** which can accomplish the equilibrium with zwitterion **A′**. Zwitterion **A** is more stable than **A′** because of the π-electrons conjugate system, so the incorporation of deuterium in γ (8%) is less than α (56%). Then, it deprotonated the pronucleophile **1a** to provide the intermediates **B** and **C**. Subsequent nucleophilic addition of them at the remote ε position resulted in the Zwitterion D that can also accomplish the equilibrium with intermediate D′. Compared with D′, PPh_3^+ exhibits a stronger electron-withdrawing ability; so zwitterion D is more stable; so the incorporation of deuterium in δ (17%) is less than β (50%). The following H shift and release of the

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547

phosphine provided the desired **3a**. The zwitterionic intermediate **A**, **D**, and their conjugated isomers can account for the deuterium distribution.

In summary, we have developed a phosphine-catalyzed remote 1,7-addition of γ-vinyl-substituted allenoates, affording a series of diene carboxylates in good to excellent yield with sole regio- and stereoselectivity. When using 1,3-dicarbonyl compounds bearing two α-hydrogens as the pronucleophiles, the corresponding double remote 1,7-addition products were obtained in moderate to good yields with an all-carbon quaternary center, which are difficult to be synthetized by traditional methods. The preliminary asymmetric version of this remote addition can give the moderate to good yields and ee values by using the chiral phosphine catalyst. The application of the corresponding diene carboxylates for the synthesis of other useful compounds were also demonstrated. The introduction of vinyl into allenoates effectively extended reaction types of phosphine-catalyzed nucleophilic additions of allenoates. Exploring more applications of this type of substrates is currently underway in our lab.[20]

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c00588.

3l-shelxl crystallographic data (CIF)

4e-R20180327B1 crystallographic data (CIF)

Experimental details, characterization data for new compounds, copies of nuclear magnetic resonance spectra and X-ray crystal structure (PDF)

■ AUTHOR INFORMATION

Corresponding Author

You Huang − *State Key Laboratory and Institute of Elemento-Organic Chemistry, College of Chemistry, Nankai University, Tianjin 300071, China;* orcid.org/0000-0002-9430-4034; Email: hyou@nankai.edu.cn

Author

Jiaxu Feng − *State Key Laboratory and Institute of Elemento-Organic Chemistry, College of Chemistry, Nankai University, Tianjin 300071, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.0c00588

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the financial support by the National Natural Science Foundation of China (no. 21871148, 21672109 and 21472097). We thank Prof. Mengchun Ye of Nankai University for helpful discussions.

■ DEDICATION

Dedicated to 100th anniversary of Nankai University and 100th anniversary of the birth of Academician Ruyu Chen.

■ REFERENCES

(1) (a) Mevers, E.; Saurí, J.; Liu, Y.; Moser, A.; Ramadhar, T. R.; Varlan, M.; Williamson, R. T.; Martin, G. E.; Clardy, J.; Homodimericin, A. A Complex Hexacyclic Fungal Metabolite. *J. Am. Chem. Soc.* **2016**, *138*, 12324−12327. (b) Steinmetz, H.; Gerth, K.; Jansen, R.; Schläger, N.; Dehn, R.; Reinecke, S.; Kirschning, A.; Müller, R.; Elansolid, A. a unique macrolide antibiotic from Chitinophaga sancti isolated as two stable atropisomers. *Angew. Chem., Int. Ed.* **2011**, *50*, 532−536. (c) Chany, A.-C.; Casarotto, V.; Schmitt, M.; Tarnus, C.; Guenin-Macé, L.; Demangel, C.; Mirguet, O.; Eustache, J.; Blanchard, N. A Diverted Total Synthesis of Mycolactone Analogues: An Insight into Buruli Ulcer Toxins. *Chem. Eur J.* **2011**, *17*, 14413−14419. (d) Paterson, I.; Paquet, T.; Dalby, S. M. Synthesis of the Macrocyclic Core of Leiodermatolide. *Org. Lett.* **2011**, *13*, 4398−4401. (e) Louis, I.; Hungerford, N. L.; Humphries, E. J.; McLeod, M. D. Enantioselective Total Synthesis of (−)-Dactylolide. *Org. Lett.* **2006**, *8*, 1117−1120. (f) Davies, S. G.; Haggitt, J. R.; Ichihara, O.; Kelly, R. J.; Leech, M. A.; Price Mortimer, A. J.; Roberts, P. M.; Smith, A. D. Asymmetric Total Synthesis of Sperabillins B and D via Lithium Amide Conjugate Addition. *Org. Biomol. Chem.* **2004**, *2*, 2630−2649. (g) Seebach, D.; Chow, H. F.; Jackson, R.; Lawson, K.; Sutter, M.; Thaisrivongs, S.; Zimmermann, J. Total Synthesis of (+)-11,11′-Di-O-Methylelaiophylidene, an Aglycon of Elaiophylin. *J. Am. Chem. Soc.* **1985**, *107*, 5292−5293.

(2) (a) Fu, K.; Zhang, J.; Lin, L.; Li, J.; Liu, X.; Feng, X.; Chiral, N. N′-Dioxide/Lanthanide(III) Complex Catalyzed Asymmetric Bisvinylogous Mukaiyama Aldol Reactions. *Org. Lett.* **2017**, *19*, 332−335. (b) Fukuda, A.; Kobayashi, Y.; Kimachi, T.; Takemoto, Y. Synthetic Studies on Macrolactin A by using a (diene)Fe(CO)3 Complex. *Tetrahedron* **2003**, *59*, 9305−9313. (c) Kim, Y.; Singer, R. A.; Carreira, E. M. Total Synthesis of Macrolactin A with Versatile Catalytic, Enantioselective Dienolate Aldol Addition Reactions. *Angew. Chem., Int. Ed.* **1998**, *37*, 1261−1263. (d) Smith, A. B.; Ott, G. R. Total Syntheses of (−)-Macrolactin A, (+)-Macrolactin E, and (−)-Macrolactinic Acid: An Exercise in Stille Cross-Coupling Chemistry. *J. Am. Chem. Soc.* **1998**, *120*, 3935−3948.

(3) (a) Denmark, S. E.; Regens, C. S.; Kobayashi, T. Total Synthesis of Papulacandin D. *J. Am. Chem. Soc.* **2007**, *129*, 2774−2776. (b) Balachari, D.; O'Doherty, G. A. Sharpless Asymmetric Dihydroxylation of 5-Aryl-2-vinylfurans: Application to the Synthesis of the Spiroketal Moiety of Papulacandin D. *Org. Lett.* **2000**, *2*, 863−866. (c) Barrett, A. G. M.; Peña, M.; Willardsen, J. A. Total Synthesis and Structural Elucidation of the Antifungal Agent Papulacandin D. *J. Org. Chem.* **1996**, *61*, 1082−1100.

(4) (a) Adamson, N. J.; Hull, E.; Malcolmson, S. J. Enantioselective Intermolecular Addition of Aliphatic Amines to Acyclic Dienes with a Pd-PHOX Catalyst. *J. Am. Chem. Soc.* **2017**, *139*, 7180−7183. (b) Lee, R. J.; Lindley, M. R.; Pritchard, G. J.; Kimber, M. C. A Biosynthetically Inspired Route to Substituted Furans Using the Appel Reaction: Total Synthesis of the Furan Fatty Acid F5. *Chem. Commun.* **2017**, *53*, 6327−6330. (c) Tao, Z.-L.; Adili, A.; Shen, H.-C.; Han, Z.-Y.; Gong, L.-Z. Catalytic Enantioselective Assembly of Homoallylic Alcohols from Dienes, Aryldiazonium Salts, and Aldehydes. *Angew. Chem., Int. Ed.* **2016**, *55*, 4322−4326. (d) Zhu, Y.; Cornwall, R. G.; Du, H.; Zhao, B.; Shi, Y. Catalytic Diamination of Olefins via N-N Bond Activation. *Acc. Chem. Res.* **2014**, *47*, 3665−3678. (e) Kundu, S.; Lyons, T. W.; Brookhart, M. Synthesis of Piperylene and Toluene via Transfer Dehydrogenation of Pentane and Pentene. *ACS Catal.* **2013**, *3*, 1768−1773. (f) McCammant, M. S.; Liao, L.; Sigman, M. S. Palladium-Catalyzed 1,4-Difunctionalization of Butadiene to form Skipped Polyenes. *J. Am. Chem. Soc.* **2013**, *135*, 4167−4170. (g) Lishchynskyi, A.; Muñiz, K. An Approach to the Regioselective Diamination of Conjugated di- and trienes. *Chem. Eur J.* **2012**, *18*, 2212−2216. (h) Ely, R. J.; Morken, J. P. Ni(0)-Catalyzed 1,4-Selective Diboration of Conjugated Dienes. *Org. Lett.* **2010**, *12*, 4348−4351. (i) Barluenga, J.; Moriel, P.; Aznar, F.; Valdés, C. A Very Simple Synthesis of Chloroalkenes and Chlorodienes by Selective Suzuki Couplings of 1,1- and 1,2-Dichloroethylene. *Adv. Synth. Catal.* **2006**, *348*, 347−353.

(5) (a) Yu, C.; Li, F.; Zhang, J.; Zhong, G. A Direct Cross-Coupling Reaction of Electron-Deficient Alkenes using an Oxidizing Directing Group. *Chem. Commun.* **2017**, *53*, 533−536. (b) Zhong, X.-M.; Cheng, G.-J.; Chen, P.; Zhang, X.; Wu, Y.-D. Mechanistic Study on Pd/Mono-N-protected Amino Acid Catalyzed Vinyl-Vinyl Coupling Reactions:

Reactivity and E/Z Selectivity. *Org. Lett.* **2016**, *18*, 5240−5243. (c) Hornillos, V.; Giannerini, M.; Vila, C.; Fañanás-Mastral, M.; Feringa, B. L. Direct Catalytic Cross-Coupling of Alkenyllithium Compounds. *Chem. Sci.* **2015**, *6*, 1394−1398. (d) Hu, X.-H.; Yang, X.-F.; Loh, T.-P. Selective Alkenylation and Hydroalkenylation of Enol Phosphates through Direct C-H Functionalization. *Angew. Chem., Int. Ed.* **2015**, *54*, 15535−15539. (e) Hu, X.-H.; Zhang, J.; Yang, X.-F.; Xu, Y.-H.; Loh, T.-P. Stereo- and Chemoselective Cross-Coupling between two Electron-Deficient Acrylates: an Efficient Route to (Z,E)-Muconate Derivatives. *J. Am. Chem. Soc.* **2015**, *137*, 3169−3172. (f) Madden, K. S.; David, S.; Knowles, J. P.; Whiting, A. Heck-Mizoroki Coupling of Vinyliodide and Applications in the Synthesis of Dienes and Trienes. *Chem. Commun.* **2015**, *51*, 11409−11412. (g) Shang, X.; Liu, Z.-Q. Transition Metal-Catalyzed C(vinyl)-C(vinyl) Bond Formation via Double C(vinyl)-H Bond Activation. *Chem. Soc. Rev.* **2013**, *42*, 3253−3260. (h) Gais, H.-J.; Rao, C. V.; Loo, R. Anionic Cross-Coupling Reaction of Alpha-Metallated Alkenyl Sulfoximines and Alkenyl Sulfoximines with Cuprates Featuring a 1,2-Metal-ate Rearrangement of Sulfoximine-substituted Higher Order Alkenyl Cuprates and an Alpha-Metallation of Alkenyl Sulfoximines by Cuprates. *Chem. Eur J.* **2008**, *14*, 6510−6528. (i) Qian, M.; Huang, Z.; Negishi, E.-i. Use of $InCl_3$ as a Cocatalyst and a $Cl_2Pd(DPEphos)-P(2\text{-Furyl})_3$ Catalyst System for One-Pot Hydrometalation−Cross-Coupling and Carbometalation−Cross-Coupling Tandem Processes. *Org. Lett.* **2004**, *6*, 1531−1534.

(6) (a) Bisceglia, J. A.; Liliana, R. O. Recent Applications of the Horner-Wadsworth-Emmons Reaction to the Synthesis of Natural Products. *Curr. Org. Chem.* **2012**, *16*, 2206−2230. (b) Wei, X.; Taylor, R. J. K. In Situ Manganese Dioxide Alcohol Oxidation−Wittig Reactions: Preparation of Bifunctional Dienyl Building Blocks. *J. Org. Chem.* **2000**, *65*, 616−620. (c) Wang, Q.; Wei, H.-x.; Schlosser, M. The Simultaneous in-situ Generation of Aldehydes and Phosphorus Ylides: A Convenient Multi-Step One-Pot Olefination Protocol. *Eur. J. Org. Chem.* **1999**, *1999*, 3263−3268. (d) Maryanoff, B. E.; Reitz, A. B. The Wittig Olefination Reaction and Modifications Involving Phosphoryl-Stabilized Carbanions. Stereochemistry, Mechanism, and Selected Synthetic Aspects. *Chem. Rev.* **1989**, *89*, 863−927.

(7) For examples of isomerization of allene, see: (a) Hampton, C. S.; Harmata, M. Mechanistic Aspects of the Phosphine-Catalyzed Isomerization of Allenic Sulfones to 2-Arylsulfonyl 1,3-Dienes. *J. Org. Chem.* **2015**, *80*, 12151−12158. (b) Hampton, C. S.; Harmata, M. Regiodivergent Synthesis of 1- and 2-Arylsulfonyl 1,3-Dienes. *Org. Lett.* **2014**, *16*, 1256−1259. (c) Dong, W.; Hu, P.; Hu, J.; Tong, X. Lewis Base-Catalyzed Divergent Isomerizations of 5-Hydroxyl-2,3-Dienoate. *Tetrahedron Lett.* **2014**, *55*, 1682−1685. (d) Xu, S.; Zhou, L.; Zeng, S.; Ma, R.; Wang, Z.; He, Z. Phosphine-Mediated Olefination between Aldehydes and Allenes: An Efficient Synthesis of Trisubstituted 1,3-Dienes with High E-Selectivity. *Org. Lett.* **2009**, *11*, 3498−3501. (e) Zhang, C.; Lu, X. Umpolung Addition Reaction of Nucleophiles to 2,3-Butadienoates Catalyzed by a Phosphine. *Synlett* **1995**, *1995*, 645−646. (f) Trost, B. M.; Kazmaier, U. Internal Redox Catalyzed by Triphenylphosphine. *J. Am. Chem. Soc.* **1992**, *114*, 7933−7935.

(8) (a) Uraguchi, D.; Shibazaki, R.; Tanaka, N.; Yamada, K.; Yoshioka, K.; Ooi, T. Catalyst-Enabled Site-Divergent Stereoselective Michael Reactions: Overriding Intrinsic Reactivity of Enynyl Carbonyl Acceptors. *Angew. Chem., Int. Ed.* **2018**, *57*, 4732−4736. (b) Woody, D.; Padarti, A.; Han, H. Asymmetric Bisvinylogous Aldolation of Aldehydes via 2-Oxonia-Cope Rearrangement Enabling Total Stereochemical Control. *Org. Lett.* **2018**, *20*, 2472−2476. (c) Gao, S.; Liu, H.; Yang, C.; Fu, Z.; Yao, H.; Lin, A. Accessing 1,3-Dienes via Palladium-Catalyzed Allylic Alkylation of Pronucleophiles with Skipped Enynes. *Org. Lett.* **2017**, *19*, 4710−4713. (d) Matovic, N. J.; Hayes, P. Y.; Penman, K.; Lehmann, R. P.; De Voss, J. J. Polyunsaturated Alkyl Amides from Echinacea: Synthesis of Diynes, Enynes, and Dienes. *J. Org. Chem.* **2011**, *76*, 4467−4481. (e) Matovic, N.; Matthias, A.; Gertsch, J.; Raduner, S.; Bone, K. M.; Lehmann, R. P.; Devoss, J. J. Stereoselective synthesis, natural Occurrence and CB_2 Receptor Binding Affinities of Alkylamides from Herbal Medicines such as Echinacea sp. *Org. Biomol. Chem.* **2007**, *5*, 169−174.

(9) (a) Guo, H.; Fan, Y. C.; Sun, Z.; Wu, Y.; Kwon, O. Phosphine Organocatalysis. *Chem. Rev.* **2018**, *118*, 10049−10293. (b) Ni, H.; Chan, W.-L.; Lu, Y. Phosphine-Catalyzed Asymmetric Organic Reactions. *Chem. Rev.* **2018**, *118*, 9344−9411. (c) Fan, Y. C.; Kwon, O. Advances in Nucleophilic Phosphine Catalysis of Alkenes, Allenes, Alkynes, and MBHADs. *Chem. Commun.* **2013**, *49*, 11588−11619. (d) Wei, Y.; Shi, M. Recent Advances in Organocatalytic Asymmetric Morita-Baylis-Hillman/Aza-Morita-Baylis-Hillman Reactions. *Chem. Rev.* **2013**, *113*, 6659−6690. (e) Basavaiah, D.; Reddy, B. S.; Badsara, S. S. Recent Contrbutions from the Baylis−Hillman Reaction to Organic Chemistry. *Chem. Rev.* **2010**, *110*, 5447−5674. (f) Wei, Y.; Shi, M. Multifunctional Chiral Phosphine Organocatalysts in Catalytic Asymmetric Morita−Baylis−Hillman and Related Reactions. *Acc. Chem. Res.* **2010**, *43*, 1005−1018. (g) Aroyan, C. E.; Miller, S. J. Enantioselective Rauhut−Currier Reactions Promoted by Protected Cysteine. *J. Am. Chem. Soc.* **2007**, *129*, 256−257. (h) Nair, V.; Menon, R. S.; Sreekanth, A. R.; Abhilash, N.; Biju, A. T. Engaging Zwitterions in Carbon−Carbon and Carbon−Nitrogen Bond-Forming Reactions: A Promising Synthetic Strategy. *Acc. Chem. Res.* **2006**, *39*, 520−530. (i) Methot, J. L.; Roush, W. R. Nucleophilic Phosphine Organocatalysis. *Adv. Synth. Catal.* **2004**, *346*, 1035−1050. (j) Lu, X.; Zhang, C.; Xu, Z. Reactions of Electron-Deficient Alkynes and Allenes under Phosphine Catalysis. *Acc. Chem. Res.* **2001**, *34*, 535−544.

(10) (a) Yoshioka, K.; Yamada, K.; Uraguchi, D.; Ooi, T. Unique Site-Selectivity Control in Asymmetric Michael Addition of Azlactone to Alkenyl Dienyl Ketones Enabled by P-spiro Chiral Iminophosphorane Catalysis. *Chem. Commun.* **2017**, *53*, 5495−5498. (b) Li, E.; Jin, H.; Jia, P.; Dong, X.; Huang, Y. Bifunctional-Phosphine-Catalyzed Sequential Annulations of Allenoates and Ketimines: Construction of Functionalized Poly-heterocycle Rings. *Angew. Chem., Int. Ed.* **2016**, *55*, 11591−11594. (c) Ziegler, D. T.; Riesgo, L.; Ikeda, T.; Fujiwara, Y.; Fu, G. C. Biphenyl-Derived Phosphepines as Chiral Nucleophilic Catalysts: Enantioselective [4+1] Annulations To Form Functionalized Cyclopentenes. *Angew. Chem., Int. Ed.* **2014**, *53*, 13183−13187. (d) Gicquel, M.; Gomez, C.; Retailleau, P.; Voituriez, A.; Marinetti, A. Synthesis of 3,3′-Spirocyclic Oxindoles via Phosphine Catalyzed [4 + 2] Cyclizations. *Org. Lett.* **2013**, *15*, 4002−4005. (e) Han, X.; Wang, Y.; Zhong, F.; Lu, Y. Enantioselective [3 + 2] Cycloaddition of Allenes to Acrylates Catalyzed by Dipeptide-Derived Phosphines: Facile Creation of Functionalized Cyclopentenes Containing Quaternary Stereogenic Centers. *J. Am. Chem. Soc.* **2011**, *133*, 1726−1729. (f) Wang, T.; Ye, S. Diastereoselective Synthesis of 6-Trifluoromethyl-5,6-dihydropyrans via Phosphine-Catalyzed [4 + 2] Annulation of α-Benzylallenoates with Ketones. *Org. Lett.* **2010**, *12*, 4168−4171. (g) Zhang, Q.; Yang, L.; Tong, X. 2-(Acetoxymethyl)buta-2,3-dienoate, a Versatile 1,4-Biselectrophile for Phosphine-Catalyzed (4 + n) Annulations with 1,n-Bisnucleophiles (n = 1, 2). *J. Am. Chem. Soc.* **2010**, *132*, 2550−2551. (h) Chung, Y. K.; Fu, G. C. Phosphine-Catalyzed Enantioselective Synthesis of Oxygen Heterocycles. *Angew. Chem., Int. Ed.* **2009**, *48*, 2225−2227. (i) Wurz, R. P.; Fu, G. C. Catalytic Asymmetric Synthesis of Piperidine Derivatives through the [4 + 2] Annulation of Imines with Allenes. *J. Am. Chem. Soc.* **2005**, *127*, 12234−12235. (j) Zhu, X.-F.; Lan, J.; Kwon, O. An Expedient Phosphine-Catalyzed [4 + 2] Annulation: Synthesis of Highly Functionalized Tetrahydropyridines. *J. Am. Chem. Soc.* **2003**, *125*, 4716−4717.

(11) (a) Wang, Z.; Xu, X.; Kwon, O. Phosphine Catalysis of Allenes with Electrophiles. *Chem. Soc. Rev.* **2014**, *43*, 2927−2940. (b) Han, X.; Yao, W.; Wang, T.; Tan, Y. R.; Yan, Z.; Kwiatkowski, J.; Lu, Y. Asymmetric Synthesis of Spiropyrazolones through Phosphine-Catalyzed [4+1] Annulation. *Angew. Chem., Int. Ed.* **2014**, *53*, 5643−5647. (c) Sun, Y.-W.; Guan, X.-Y.; Shi, M. Synthesis of Functionalized Chromans by PBu3-Catalyzed Reactions of Salicylaldimines and Salicylaldehydes with Allenic Ester. *Org. Lett.* **2010**, *12*, 5664−5667. (d) Cowen, B. J.; Miller, S. J. Enantioselective Catalysis and Complexity Generation from Allenoates. *Chem. Soc. Rev.* **2009**, *38*, 3102−3116. (e) Guo, H.; Xu, Q.; Kwon, O. Phosphine-Promoted [3 + 3] Annulations of Aziridines with Allenoates: Facile Entry into Highly Functionalized Tetrahydropyridines. *J. Am. Chem. Soc.* **2009**, *131*, 6318−6319. (f) Tran, Y. S.; Kwon, O. Phosphine-Catalyzed [4 + 2]

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547

Annulation: Synthesis of Cyclohexenes. *J. Am. Chem. Soc.* **2007**, *129*, 12632−12633. (g) Henry, C. E.; Kwon, O. Phosphine-Catalyzed Synthesis of Highly Functionalized Coumarins. *Org. Lett.* **2007**, *9*, 3069−3072. (h) Ye, L.-W.; Sun, X.-L.; Wang, Q.-G.; Tang, Y. Phosphine-Catalyzed Intramolecular Formal [3+2] Cycloaddition for Highly Diastereoselective Synthesis of Bicyclo[n.3.0] Compounds. *Angew. Chem., Int. Ed.* **2007**, *46*, 5951−5954. (i) Wang, J.-C.; Ng, S.-S.; Krische, M. J. Catalytic Diastereoselective Synthesis of Diquinanes from Acyclic Precursors. *J. Am. Chem. Soc.* **2003**, *125*, 3682−3683. (j) Zhang, C.; Lu, X. Phosphine-Catalyzed Cycloaddition of 2,3-Butadienoates or 2-Butynoates with Electron-Deficient Olefins. A Novel [3 + 2] Annulation Approach to Cyclopentenes. *J. Org. Chem.* **1995**, *60*, 2906−2908.

(12) (a) Wang, H.; Guo, C. Enantioselective γ-Addition of Pyrazole and Imidazole Heterocycles to Allenoates Catalyzed by Chiral Phosphine. *Angew. Chem., Int. Ed.* **2019**, *58*, 2854−2858. (b) Wang, T.; Yu, Z.; Hoon, D. L.; Phee, C. Y.; Lan, Y.; Lu, Y. Regiodivergent Enantioselective γ-Additions of Oxazolones to 2,3-Butadienoates Catalyzed by Phosphines: Synthesis of alpha,alpha-Disubstituted alpha-Amino Acids and N,O-Acetal Derivatives. *J. Am. Chem. Soc.* **2016**, *138*, 265−271. (c) Wang, T.; Yu, Z.; Hoon, D. L.; Huang, K.-W.; Lan, Y.; Lu, Y. Highly Enantioselective Construction of Tertiary Thioethers and Alcohols via Phosphine-Catalyzed Asymmetric γ-Addition Reactions of 5H-Thiazol-4-Ones And 5H-Oxazol-4-Ones: Scope and Mechanistic Understandings. *Chem. Sci.* **2015**, *6*, 4912−4922. (d) Wang, T.; Yao, W.; Zhong, F.; Pang, G. H.; Lu, Y. Phosphine-Catalyzed Enantioselective γ-addition of 3-Substituted Oxindoles to 2,3-Butadienoates and 2-Butynoates: use of Prochiral Nucleophiles. *Angew. Chem., Int. Ed.* **2014**, *53*, 2964−2968. (e) Smith, S. W.; Fu, G. C. Asymmetric Carbon−Carbon Bond Formation γ to a Carbonyl Group: Phosphine-Catalyzed Addition of Nitromethane to Allenes. *J. Am. Chem. Soc.* **2009**, *131*, 14231−14233.

(13) (a) Trost, B. M.; Li, C.-J. Novel "Umpolung" in C-C Bond Formation Catalyzed by Triphenylphosphine. *J. Am. Chem. Soc.* **1994**, *116*, 3167−3168. (b) Trost, B. M.; Li, C.-J. Phosphine-Catalyzed Isomerization-Addition of Oxygen Nucleophiles to 2-Alkynoates. *J. Am. Chem. Soc.* **1994**, *116*, 10819−10820.

(14) (a) Lundgren, R. J.; Wilsily, A.; Marion, N.; Ma, C.; Chung, Y. K.; Fu, G. C. Catalytic Asymmetric C-N Bond Formation: Phosphine-Catalyzed Intra- and Intermolecular Γ-Addition of Nitrogen Nucleophiles to Allenoates and Alkynoates. *Angew. Chem., Int. Ed.* **2013**, *52*, 2525−2528. (b) Andrews, I. P.; Blank, B. R.; Kwon, O. Phosphine-Catalyzed Intramolecular Γ-Umpolung Addition of Alpha-Aminoalkylallenic Esters: Facile Synthesis of 3-Carbethoxy-2-Alkyl-3-Pyrrolines. *Chem. Commun.* **2012**, *48*, 5373−5375. (c) Guan, X.-Y.; Wei, Y.; Shi, M. Phosphine-Catalyzed Tandem Reaction of Allenoates with Nitroalkenes. *Org. Lett.* **2010**, *12*, 5024−5027. (d) Sinisi, R.; Sun, J.; Fu, G. C. Phosphine-Catalyzed Asymmetric Additions of Malonate Esters to γ-Substituted Allenoates and Allenamides. *Proc. Natl. Acad. Sci. U.S.A* **2010**, *107*, 20652−20654. (e) Lu, C.; Lu, X. Tandem Reactions to Construct Heterocycles via Phosphine-Catalyzed Umpolung Addition and Intramolecular Conjugate Addition. *Org. Lett.* **2002**, *4*, 4677−4679. (f) Chen, Z.; Zhu, G.; Jiang, Q.; Xiao, D.; Cao, P.; Zhang, X. Asymmetric Formation of Quaternary Carbon Centers Catalyzed by Novel Chiral 2,5-Dialkyl-7-phenyl-7-phosphabicyclo[2.2.1]heptanes. *J. Org. Chem.* **1998**, *63*, 5631−5635.

(15) (a) Huang, Z.; Yang, X.; Yang, F.; Lu, T.; Zhou, Q. Phosphine-Catalyzed Domino β/γ-Additions of Benzofuranones with Allenoates: A Method for Unsymmetrical 3,3-Disubstituted Benzofuranones. *Org. Lett.* **2017**, *19*, 3524−3527. (b) Gandi, V. R.; Lu, Y. Phosphine-Catalyzed Regioselective Michael Addition to Allenoates. *Chem. Commun.* **2015**, *51*, 16188−16190. (c) Li, E.; Xie, P.; Yang, L.; Liang, L.; Huang, Y. Tuning Catalysts to Tune the Products: Phosphine-Catalyzed aza-Michael Addition Reaction of Hydrazones with Allenoates. *Chem. Asian J.* **2013**, *8*, 603−610. (d) Szeto, J.; Sriramurthy, V.; Kwon, O. Phosphine-Initiated General Base Catalysis: Facile Access to Benzannulated 1,3-Diheteroatom Five-Membered Rings via Double-Michael Reactions of Allenes. *Org. Lett.* **2011**, *13*, 5420−5423.

(16) (a) Guan, X.-Y.; Wei, Y.; Shi, M. Phosphane-Catalyzed Umpolung Addition Reaction of Nucleophiles to Ethyl 2-Methyl-2,3-butadienoate. *Eur. J. Org. Chem.* **2011**, *2011*, 2673−2677. (b) Martin, T. J.; Vakhshori, V. G.; Tran, Y. S.; Kwon, O. Phosphine-Catalyzed β′-Umpolung Addition of Nucleophiles to Activated α-Alkyl Allenes. *Org. Lett.* **2011**, *13*, 2586−2589.

(17) (a) Meng, X.; Huang, Y.; Zhao, H.; Xie, P.; Ma, J.; Chen, R. PPh3-Catalyzed Domino Reaction: A Facile Method for the Synthesis of Chroman Derivatives. *Org. Lett.* **2009**, *11*, 991−994. (b) Zhang, Q.; Zhu, Y.; Jin, H.; Huang, Y. A Phosphine Mediated Sequential Annulation Process of 2-Tosylaminochalcones with MBH Carbonates to Construct Functionalized aza-Benzobicyclo[4.3.0] Derivatives. *Chem. Commun.* **2017**, *53*, 3974−3977. (c) Li, E.; Chang, M.; Liang, L.; Huang, Y. Divergent Phosphine-Catalyzed [2+4] or [3+2] Cycloaddition Reactions of γ-Substituted Allenoates with Oxadienes. *Eur. J. Org. Chem.* **2015**, *2015*, 710−714. (d) Li, E.; Huang, Y. Phosphine-Catalyzed Domino Reaction: A Novel Sequential [2+3] and [3+2] Annulation Reaction of γ-Substituent Allenoates to Construct Bicyclic[3, 3, 0]Octene Derivatives. *Chem. Commun.* **2014**, *50*, 948−950. (e) Zheng, J.; Huang, Y.; Li, Z. Phosphine-Catalyzed Sequential Annulation Domino Reaction: Rapid Construction of Bicyclo[4.1.0]-Heptene Skeletons. *Chem. Commun.* **2014**, *50*, 5710−5713. (f) Li, N.; Jia, P.; Huang, Y. Facile access to highly functionalized hydroisoquinoline derivatives via phosphinecatalyzed sequential [3+3]/[3+3] annulation. *Chem. Commun.* **2019**, *55*, 10976−10979.

(18) CCDC 1850634 (3l) and CCDC 1850821 (4e) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

(19) Deng, H.-P.; Wei, Y.; Shi, M. Enantioselective Synthesis of Highly Functionalized Trifluoromethyl-Bearing Cyclopentenes: Asymmetric [3+2] Annulation of Morita−Baylis−Hillman Carbonates with Trifluoroethylidenemalonates Catalyzed by Multifunctional Thiourea-Phosphines. *Adv. Synth. Catal.* **2012**, *354*, 783−789.

(20) (a) Feng, J.; Huang, Y. Phosphine-Catalyzed (3+2)/(2+3) Sequential Annulation by Triple Nucleophilic Addition Reaction of γ-Vinyl Allenoates. *Chem. Commun.* **2019**, *55*, 14011−14014. (b) Feng, J.; Chen, Y.; Qin, W.; Huang, Y. Phosphine-Catalyzed (3 + 2)/(3 + 2) Sequential Annulation of γ-Vinyl Allenoates: Access to Fused Carbocycles Org. *Lett* **2020**, *22*, 433−437.

https://dx.doi.org/10.1021/acscatal.0c00588
ACS Catal. 2020, 10, 3541−3547